

# Manual

## HYDRA Interfacing Module for SAP MM SAP-MMMOB 3.0

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# 1 HYDRA Interfacing Module to SAP MM

## Summary

### Fields of application

The function package provides a communication container (IDoc) by which consumptions and goods receipt recorded in the MES can be transferred to SAP ECC. Data are transferred in the SAP standard format.

### Implementation notes

You use the function package if you wish:

- to record MES consumptions (with and without relation to batches) and upload them to SAP
- to upload material generated in MES as goods receipt in SAP
- to perform the upload in the SAP standard format and not according to the customer's requirements

### Integration

The function package is also integrated in the following function packages:

- MPL-MBV – material and inventory management
- TRT-CLV – batch data management
- TRT-CLA – batch data processing

### Features

- Interfacing module to connect HYDRA to SAP MM using the MM-MOB interface. Upload of goods movements using the SAP standard IDoc WMMBID02 (segments E2MBXYH, E2MBXYI):
- Transfer of goods receipts from the production (SAP movement type 101)
- Transfer of goods issues from the production (SAP movement type 261)
- Prerequisite: HYDRA modules (e.g. MPL) generating goods movements are used and technical details are coordinated (chargeable customization service).



Prerequisites: HYDRA modules (e.g. MPL, TRT) generating goods movements are in use.

## 2 Mapping of MM-MOB in HYDRA

### Summary

The function package SAP-MMMOB is based on [goods movements](#) generated by other function packages (e.g. MPL, TRT). Goods receipts as well as goods issues are considered in this context.

The interface itself does not perform any further checks. Data consistency has to be ensured by the generating applications.

### Prerequisite

The MES material type has to be configured as subject to uploads for the interface to make sure that goods movements can be provided.

### 3 Upload of Goods Movements HYDRA --> SAP

#### 3.1 Goods issues

Material withdrawals (consumption) recorded in HYDRA are transferred to SAP.

HYDRA provides the SAP system with the data in cyclic intervals. The SAP standard IDoc WMMBID02 of the message type WMMBXY is used for this purpose.

|               |                           |        |
|---------------|---------------------------|--------|
| Message type: | WMMBXY                    |        |
| IDOC type     | WMMBID02                  |        |
| Segments:     | E2MBXYH (header record)   | single |
|               | └ E2MBXYI (detail record) | single |

#### Transaction types

| Transaction type | Description/usage                                              |
|------------------|----------------------------------------------------------------|
| 261              | Goods issue for production order (consumption)                 |
| 262              | Cancellation of goods issue for production order (consumption) |

#### Data structure (E2MBXYH)

| Field | T    | L  | D | Designation                                           | Usage in HYDRA                    |
|-------|------|----|---|-------------------------------------------------------|-----------------------------------|
| BLDAT | DATE | 8  |   | Document date in document                             | Date when consumption was entered |
| BUDAT | DATE | 8  |   | Posting date in document                              | Date of material movement         |
| XBLNR | CHAR | 16 |   | Reference document number                             | Reference of material movement    |
| BKTX  | CHAR | 25 |   | Document header text                                  | Not used                          |
| FRBNR | CHAR | 16 |   | Number of the bill of lading at time of goods receipt | Not used                          |
| XABLN | CHAR | 10 |   | Goods receipt/issue slip number                       | Not used                          |
| TCODE | CHAR | 4  |   | Session: Current transaction code                     | Fixed "MB1A"                      |
| USNAM | CHAR | 12 |   | User name (release 4.0)                               | Not used                          |
| VBUND | CHAR | 6  |   | Society number (release 4.0)                          | Not used                          |

## Data structure (E2MBXYI)

| Field | T    | L  | D | Designation                           | Usage in HYDRA                                                |
|-------|------|----|---|---------------------------------------|---------------------------------------------------------------|
| BEAKZ | CHAR | 1  |   | Indicator: line already edited        | Not used                                                      |
| XSTOB | CHAR | 1  |   | Flag: Reverse posting                 | Not used                                                      |
| MATNR | CHAR | 18 |   | Material number                       | Material number of the component to be withdrawn              |
| WERKS | CHAR | 4  |   | Plant                                 | Factory of the production order                               |
| LGORT | CHAR | 4  |   | Storage location                      | Storage location from user field 46 of the component list     |
| CHARG | CHAR | 10 |   | Batch number                          | Recorded batch number                                         |
| BWART | CHAR | 3  |   | Movement type (inventory management)  | "261" for goods issues<br>"262" cancellation for goods issue  |
| INSMK | CHAR | 1  |   | Stock type                            | Not used                                                      |
| SOBKZ | CHAR | 1  |   | Special stock indicator               | Not used                                                      |
| KZVBR | CHAR | 1  |   | Indicator: consumption posting        | Not used                                                      |
| LIFNR | CHAR | 10 |   | Vendor account number                 | Not used                                                      |
| KUNNR | CHAR | 10 |   | Customer number                       | Not used                                                      |
| KDAUF | CHAR | 10 |   | Sales order number                    | Not used                                                      |
| KDPOS | CHAR | 6  |   | Item number in customer order         | Not used                                                      |
| KDEIN | CHAR | 4  |   | Scheduling of customer order          | Not used                                                      |
| SHKZG | CHAR | 1  |   | Debit and credit indicator            | Not used                                                      |
| WAERS | CHAR | 5  |   | Currency key                          | Not used                                                      |
| DMBTR | CHAR | 15 |   | Amount in local currency              | Not used                                                      |
| BWTAR | CHAR | 10 |   | Valuation type                        | Not used                                                      |
| ERFMG | QUAN | 13 | 3 | Quantity in unit of entry             | Quantity recorded as consumption                              |
| ERFME | CHAR | 3  |   | Unit of entry                         | Quantity unit for consumption quantity                        |
| BPMNG | QUAN | 13 | 3 | Quantity in order price quantity unit | Not used                                                      |
| BPRME | CHAR | 3  |   | Order price quantity unit             | Not used                                                      |
| EBELN | CHAR | 10 |   | Purchasing document number            | Not used                                                      |
| EBELP | CHAR | 5  |   | Item number of purchasing document    | Not used                                                      |
| ELIKZ | CHAR | 1  |   | "Delivery completed" indicator        | Not used                                                      |
| SGTXT | CHAR | 50 |   | Item text                             | Not used                                                      |
| WEMPF | CHAR | 12 |   | Recipient of the goods                | Not used                                                      |
| ABLAD | CHAR | 25 |   | Place of unloading                    | Not used                                                      |
| KOSTL | CHAR | 10 |   | Cost center                           | Not used                                                      |
| AUFNR | CHAR | 12 |   | Order number                          | SAP order number for which material has been withdrawn        |
| ANLN1 | CHAR | 12 |   | Asset main number                     | Not used                                                      |
| ANLN2 | CHAR | 4  |   | Asset sub-number                      | Not used                                                      |
| RSNUM | CHAR | 10 |   | Reservation number                    | Number of the reservation from user field 45 of the component |

| Field      | T    | L  | D | Designation                                         | Usage in HYDRA                                                     |
|------------|------|----|---|-----------------------------------------------------|--------------------------------------------------------------------|
| RSPOS      | CHAR | 4  |   | Item number of the reservation                      | Item number of the reservation from user field 47 of the component |
| KZEAR      | CHAR | 1  |   | Indicator: final issue                              | Not used                                                           |
| UMMAT      | CHAR | 18 |   | Receiving/issuing material                          | Not used                                                           |
| UMWRK      | CHAR | 4  |   | Receiving/issuing plant                             | Not used                                                           |
| UMLGO      | CHAR | 4  |   | Receiving/issuing storage location                  | Not used                                                           |
| UMCHA      | CHAR | 10 |   | Receiving/issuing batch                             | Not used                                                           |
| KZBEW      | CHAR | 1  |   | Movement indicator                                  | Not used                                                           |
| WEUNB      | CHAR | 1  |   | Indicator: goods receipt                            | Not used                                                           |
| LGNUM      | CHAR | 3  |   | Warehouse number                                    | Not used                                                           |
| LGTYP      | CHAR | 3  |   | Storage type                                        | Not used                                                           |
| LGPLA      | CHAR | 10 |   | Storage bin                                         | Not used                                                           |
| GRUND      | CHAR | 4  |   | Indicator: transaction reason                       | Not used                                                           |
| EVERS      | CHAR | 2  |   | Shipping instructions                               | Not used                                                           |
| EVERE      | CHAR | 2  |   | Complying with shipping instructions                | Not used                                                           |
| IMKEY      | CHAR | 8  |   | Internal key for real estate property               | Not used                                                           |
| KSTRG      | CHAR | 12 |   | Cost object                                         | Not used                                                           |
| PAOBJNR    | CHAR | 10 |   | Number for business segment                         | Not used                                                           |
| PRCTR      | CHAR | 10 |   | Profit center                                       | Not used                                                           |
| PS_PSP_PNR | CHAR | 8  |   | Planning element of project structure               | Not used                                                           |
| NPLNR      | CHAR | 12 |   | Network number for account assignment               | Not used                                                           |
| AUFPL      | CHAR | 10 |   | Planning number for transactions in the order       | Not used                                                           |
| APLZL      | CHAR | 8  |   | Continuous counter                                  | Not used                                                           |
| AUFPS      | CHAR | 4  |   | Number of order item                                | Not used                                                           |
| VPTNR      | CHAR | 10 |   | Partner account number                              | Not used                                                           |
| FIPOS      | CHAR | 14 |   | Commitment item                                     | Not used                                                           |
| GSBER      | CHAR | 4  |   | Business area                                       | Not used                                                           |
| BSTMG      | QUAN | 13 | 3 | Goods receipt quantity in order unit                | Not used                                                           |
| BSTME      | CHAR | 3  |   | Order unit                                          | Not used                                                           |
| EXBWR      | QUAN | 13 | 3 | Posting amount in local currency entered externally | Not used                                                           |
| KONTO      | CHAR | 10 |   | G/L account number                                  | Not used                                                           |
| RSHKZ      | CHAR | 1  |   | Debit and credit indicator                          | Fixed "H"                                                          |
| BDMNG      | QUAN | 13 | 3 | Requirement quantity                                | Not used                                                           |
| ENMNG      | QUAN | 13 | 3 | Issued quantity                                     | Not used                                                           |
| QPLOS      | CHAR | 12 |   | Inspection batch number                             | Not used                                                           |
| UMZST      | CHAR | 1  |   | Status of receiving batch                           | Not used                                                           |
| UMZUS      | CHAR | 1  |   | Status key of transfer batch                        | Not used                                                           |
| UMBAR      | CHAR | 10 |   | Valuation type of transfer batch                    | Not used                                                           |
| UMSOK      | CHAR | 1  |   | Special stock indicator                             | Not used                                                           |
| LFBJA      | CHAR | 4  |   | Fiscal year of a reference document                 | Not used                                                           |
| LFBNR      | CHAR | 10 |   | Document number of a reference document             | Not used                                                           |
| LFPOS      | CHAR | 4  |   | Item in a reference document                        | Not used                                                           |
| SJAHR      | CHAR | 4  |   | Material document year                              | Not used                                                           |



| Field    | T    | L  | D | Designation                                                  | Usage in HYDRA |
|----------|------|----|---|--------------------------------------------------------------|----------------|
| SMBLN    | CHAR | 10 |   | Number of a material document                                | Not used       |
| SMBLP    | CHAR | 4  |   | Item in the material document                                | Not used       |
| EXVKW    | CHAR | 15 |   | Sales value specified externally in local currency           | Not used       |
| QM_ZUSTD | CHAR | 1  |   | Batch status with status changed in QM (internal)            | Not used       |
| POSNR    | CHAR | 6  |   | Delivery item for third-party system                         | Not used       |
| VBELN    | CHAR | 10 |   | Delivery                                                     | Not used       |
| QM_UMZST | CHAR | 1  |   | Status of received batch when status changed in QM (intern.) | Not used       |
| BWLVS    | CHAR | 3  |   | Movement type for warehouse management system                | Not used       |
| UMREZ    | CHAR | 5  |   | Numerator for converting to base unit of measure             | Not used       |
| UMREN    | CHAR | 5  |   | Denominator for conversion to base unit of measure           | Not used       |
| VFDAT    | CHAR | 8  |   | Expiration date or best-before date                          | Not used       |

## 3.2 Goods receipts

Goods receipts recorded in HYDRA (for produced material) are transferred to SAP.

HYDRA provides the SAP system with the data in cyclic intervals. The SAP standard IDoc WMMBID02 of the message type WMMBXY is used for this purpose.

|               |                           |        |
|---------------|---------------------------|--------|
| Message type: | WMMBXY                    |        |
| IDOC type     | WMMBID02                  |        |
| Segments:     | E2MBXYH (header record)   | single |
|               | └ E2MBXYI (detail record) | single |

## Transaction types

| Transaction type | Description/usage                               |
|------------------|-------------------------------------------------|
| 101              | Goods receipt for production order              |
| 102              | Goods receipt cancellation for production order |

## Data structure (E2MBXYH)

| Field | T    | L | D | Designation               | Usage in HYDRA    |
|-------|------|---|---|---------------------------|-------------------|
| BLDAT | DATE | 8 |   | Document date in document | Date of the entry |

| Field | T    | L  | D | Designation                                           | Usage in HYDRA                 |
|-------|------|----|---|-------------------------------------------------------|--------------------------------|
| BUDAT | DATE | 8  |   | Posting date in document                              | Date of material movement      |
| XBLNR | CHAR | 16 |   | Reference document number                             | Reference of material movement |
| BKTX  | CHAR | 25 |   | Document header text                                  | Not used                       |
| FRBNR | CHAR | 16 |   | Number of the bill of lading at time of goods receipt | Not used                       |
| XABLN | CHAR | 10 |   | Goods receipt/issue slip number                       | Not used                       |
| TCODE | CHAR | 4  |   | Session: Current transaction code                     | Fixed "MB31"                   |
| USNAM | CHAR | 12 |   | User name (release 4.0)                               | Not used                       |
| VBUND | CHAR | 6  |   | Society number (release 4.0)                          | Not used                       |

### Data structure (E2MBXYI)

| Field | T    | L  | D | Designation                          | Usage in HYDRA                                               |
|-------|------|----|---|--------------------------------------|--------------------------------------------------------------|
| BEAKZ | CHAR | 1  |   | Indicator: line already edited       | Not used                                                     |
| XSTOB | CHAR | 1  |   | Flag: Reverse posting                | Not used                                                     |
| MATNR | CHAR | 18 |   | Material number                      | Material number of the produced material                     |
| WERKS | CHAR | 4  |   | Plant                                | Factory of the production order producing the material       |
| LGORT | CHAR | 4  |   | Storage location                     | Storage location from user field 45 of the order header      |
| CHARG | CHAR | 10 |   | Batch number                         | Recorded batch number                                        |
| BWART | CHAR | 3  |   | Movement type (inventory management) | "101" for goods receipts<br>"102" goods receipt cancellation |
| INSMK | CHAR | 1  |   | Stock type                           | Not used                                                     |
| SOBKZ | CHAR | 1  |   | Special stock indicator              | Not used                                                     |
| KZVBR | CHAR | 1  |   | Indicator: consumption posting       | Not used                                                     |
| LIFNR | CHAR | 10 |   | Vendor account number                | Not used                                                     |
| KUNNR | CHAR | 10 |   | Customer number                      | Not used                                                     |
| KDAUF | CHAR | 10 |   | Sales order number                   | Not used                                                     |
| KDPOS | CHAR | 6  |   | Item number in customer order        | Not used                                                     |
| KDEIN | CHAR | 4  |   | Scheduling of customer order         | Not used                                                     |
| SHKZG | CHAR | 1  |   | Debit and credit indicator           | Not used                                                     |
| WAERS | CHAR | 5  |   | Currency key                         | Not used                                                     |
| DMBTR | CHAR | 15 |   | Amount in local currency             | Not used                                                     |
| BWTAR | CHAR | 10 |   | Valuation type                       | Not used                                                     |
| ERFMG | QUAN | 13 | 3 | Quantity in unit of entry            | Recorded quantity                                            |
| ERFME | CHAR | 3  |   | Unit of entry                        | Quantity unit                                                |

| Field      | T    | L  | D | Designation                                         | Usage in HYDRA                                                   |
|------------|------|----|---|-----------------------------------------------------|------------------------------------------------------------------|
| BPMNG      | QUAN | 13 | 3 | Quantity in order price quantity unit               | Not used                                                         |
| BPRME      | CHAR | 3  |   | Order price quantity unit                           | Not used                                                         |
| EBELN      | CHAR | 10 |   | Purchasing document number                          | Not used                                                         |
| EBELP      | CHAR | 5  |   | Item number of purchasing document                  | Not used                                                         |
| ELIKZ      | CHAR | 1  |   | "Delivery completed" indicator                      | Not used                                                         |
| SGTXT      | CHAR | 50 |   | Item text                                           | Not used                                                         |
| WEMPF      | CHAR | 12 |   | Recipient of the goods                              | Not used                                                         |
| ABLAD      | CHAR | 25 |   | Place of unloading                                  | Not used                                                         |
| KOSTL      | CHAR | 10 |   | Cost center                                         | Not used                                                         |
| AUFNR      | CHAR | 12 |   | Order number                                        | SAP order number for which material has been received / produced |
| ANLN1      | CHAR | 12 |   | Asset main number                                   | Not used                                                         |
| ANLN2      | CHAR | 4  |   | Asset sub-number                                    | Not used                                                         |
| RSNUM      | CHAR | 10 |   | Reservation number                                  | Not used                                                         |
| RSPOS      | CHAR | 4  |   | Item number of the reservation                      | Not used                                                         |
| KZEAR      | CHAR | 1  |   | Indicator: final issue                              | Not used                                                         |
| UMMAT      | CHAR | 18 |   | Receiving/issuing material                          | Not used                                                         |
| UMWRK      | CHAR | 4  |   | Receiving/issuing plant                             | Not used                                                         |
| UMLGO      | CHAR | 4  |   | Receiving/issuing storage location                  | Not used                                                         |
| UMCHA      | CHAR | 10 |   | Receiving/issuing batch                             | Not used                                                         |
| KZBEW      | CHAR | 1  |   | Movement indicator                                  | Fixed "F"                                                        |
| WEUNB      | CHAR | 1  |   | Indicator: goods receipt                            | Not used                                                         |
| LGNUM      | CHAR | 3  |   | Warehouse number                                    | Not used                                                         |
| LGTYP      | CHAR | 3  |   | Storage type                                        | Not used                                                         |
| LGPLA      | CHAR | 10 |   | Storage bin                                         | Not used                                                         |
| GRUND      | CHAR | 4  |   | Indicator: transaction reason                       | Not used                                                         |
| EVERS      | CHAR | 2  |   | Shipping instructions                               | Not used                                                         |
| EVERE      | CHAR | 2  |   | Complying with shipping instructions                | Not used                                                         |
| IMKEY      | CHAR | 8  |   | Internal key for real estate property               | Not used                                                         |
| KSTRG      | CHAR | 12 |   | Cost object                                         | Not used                                                         |
| PAOBJNR    | CHAR | 10 |   | Number for business segment                         | Not used                                                         |
| PRCTR      | CHAR | 10 |   | Profit center                                       | Not used                                                         |
| PS_PSP_PNR | CHAR | 8  |   | Planning element of project structure               | Not used                                                         |
| NPLNR      | CHAR | 12 |   | Network number for account assignment               | Not used                                                         |
| AUFPL      | CHAR | 10 |   | Planning number for transactions in the order       | Not used                                                         |
| APLZL      | CHAR | 8  |   | Continuous counter                                  | Not used                                                         |
| AUFPS      | CHAR | 4  |   | Number of order item                                | Not used                                                         |
| VPTNR      | CHAR | 10 |   | Partner account number                              | Not used                                                         |
| FIPOS      | CHAR | 14 |   | Commitment item                                     | Not used                                                         |
| GSBER      | CHAR | 4  |   | Business area                                       | Not used                                                         |
| BSTMG      | QUAN | 13 | 3 | Goods receipt quantity in order unit                | Not used                                                         |
| BSTME      | CHAR | 3  |   | Order unit                                          | Not used                                                         |
| EXBWR      | QUAN | 13 | 3 | Posting amount in local currency entered externally | Not used                                                         |
| KONTO      | CHAR | 10 |   | G/L account number                                  | Not used                                                         |

| Field    | T    | L  | D | Designation                                                  | Usage in HYDRA |
|----------|------|----|---|--------------------------------------------------------------|----------------|
| RSHKZ    | CHAR | 1  |   | Debit and credit indicator                                   | Fixed "S"      |
| BDMNG    | QUAN | 13 | 3 | Requirement quantity                                         | Not used       |
| ENMNG    | QUAN | 13 | 3 | Issued quantity                                              | Not used       |
| QPLOS    | CHAR | 12 |   | Inspection batch number                                      | Not used       |
| UMZST    | CHAR | 1  |   | Status of receiving batch                                    | Not used       |
| UMZUS    | CHAR | 1  |   | Status key of transfer batch                                 | Not used       |
| UMBAR    | CHAR | 10 |   | Valuation type of transfer batch                             | Not used       |
| UMSOK    | CHAR | 1  |   | Special stock indicator                                      | Not used       |
| LFBJA    | CHAR | 4  |   | Fiscal year of a reference document                          | Not used       |
| LFBNR    | CHAR | 10 |   | Document number of a reference document                      | Not used       |
| LFPOS    | CHAR | 4  |   | Item in a reference document                                 | Not used       |
| SJAHR    | CHAR | 4  |   | Material document year                                       | Not used       |
| SMBLN    | CHAR | 10 |   | Number of a material document                                | Not used       |
| SMBLP    | CHAR | 4  |   | Item in the material document                                | Not used       |
| EXVKW    | CHAR | 15 |   | Sales value specified externally in local currency           | Not used       |
| QM_ZUSTD | CHAR | 1  |   | Batch status with status changed in QM (internal)            | Not used       |
| POSNR    | CHAR | 6  |   | Delivery item for third-party system                         | Not used       |
| VBELN    | CHAR | 10 |   | Delivery                                                     | Not used       |
| QM_UMZST | CHAR | 1  |   | Status of received batch when status changed in QM (intern.) | Not used       |
| BWLVS    | CHAR | 3  |   | Movement type for warehouse management system                | Not used       |
| UMREZ    | CHAR | 5  |   | Numerator for converting to base unit of measure             | Not used       |
| UMREN    | CHAR | 5  |   | Denominator for conversion to base unit of measure           | Not used       |
| VFDAT    | CHAR | 8  |   | Expiration date or best-before date                          | Not used       |

## 4 Application-Relevant Settings in HYDRA

### Maintenance of the HYDRA distribution model - outbound processing

Use the [HYDRA distribution model](#) to maintain entries for HYDRA outbound processing:

| Parameter name                 | Value                                 |
|--------------------------------|---------------------------------------|
| For the upload of time tickets |                                       |
| Message type                   | WMMBXY                                |
| Description                    | MM-MOB - Upload of material movements |
| IDOC type                      | WMMBID02                              |
| Storage duration               | 10                                    |
| Log. target system             | Created logical system                |
| Segment name 1                 | E2MBXYH                               |

### Maintenance of the HYDRA Scheduler

Use the [HYDRA Scheduler](#) to plan jobs for outbound processing:

| Parameter name                                                           | Value                                                            |
|--------------------------------------------------------------------------|------------------------------------------------------------------|
| Transfer of goods movements to <a href="#">MLE outbound transactions</a> |                                                                  |
| Product key                                                              | SAP-MMMOB                                                        |
| License key                                                              | SAP-MMMOB                                                        |
| Command (Windows):                                                       | sh.exe ./myerprck.scr /MESTYP=WMMBXY                             |
| Command (Unix):                                                          | ./myerprck.scr /MESTYP=WMMBXY                                    |
| Comment:                                                                 | SAP-MMMOB: Upload of goods movements → MLE outbound transactions |
| Interval                                                                 | 5                                                                |

| Parameter name                                                                   | Value                                                                    |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Upload of confirmations from <a href="#">MLE outbound transactions</a> to SAP MM |                                                                          |
| Product key                                                                      | SAP-MMMOB                                                                |
| License key                                                                      | SAP-MMMOB                                                                |
| Command (Windows):                                                               | sh.exe ./.hysapupl.scr /UPLSEGNAM=E2MBXYH /SINGLE_IDOC /SUBLEVEL=2       |
| Command (Unix):                                                                  | ./hysapupl.scr /UPLSEGNAM=E2MBXYH /SINGLE_IDOC /SUBLEVEL=2               |
| Comment:                                                                         | SAP-MMMOB - Upload of goods movements MLE outbound transactions → SAP MM |
| Interval                                                                         | 5                                                                        |



Please make sure that the upload program myerprck.exe/out has not been started for the message types ZWAU (goods issues in the HYDRA format) and ZWEI (goods receipts in the HYDRA format) using the script myerprck.scr, since it cannot be operated with SAP-MMMOB at the same time.

## Maintenance of the HYDRA material type

In many cases, the HYDRA material type is relevant for the interface and needs to be activated to be able to transfer goods movements. For this reason, enable the flag “goods movements → transfer to interface” for the [HYDRA material type](#).



If the HYDRA material type is not available as application, the flag can also directly be set using the database:

```
update hz_typen set we_ext_kz = ,J' where hz_typ = '<Material type for which the flag is to be set>';
```

## 5 Application-Relevant Settings in SAP

### Maintenance of the SAP partner agreement – inbound processing

Maintain the following settings for inbound processing in the partner agreement in SAP (WE20):

| Parameter name   | Value                  |
|------------------|------------------------|
| Partner number   | Created logical system |
| Partner type     | LS                     |
| Message type     | WMMBXY                 |
| Transaction code | WMMB                   |

### Maintenance of the SAP distribution model – inbound processing

| Parameter name         | Value                                |
|------------------------|--------------------------------------|
| To upload time tickets |                                      |
| Model view             | Created model view                   |
| Sender / Client        | Logical system for the sender system |
| Recipient/ server      | Logical system of the client         |
| Message type           | WMMBXY                               |

### Relevant transactions

| Transaction | Meaning                       | Note |
|-------------|-------------------------------|------|
| MB51        | Display of material documents |      |

## Possible modifications

If the standard SAP inbound processing does not meet the requirements, it can be enhanced by numerous [modifications](#).

| Modification | Meaning                                                                                   | Note |
|--------------|-------------------------------------------------------------------------------------------|------|
| MWMIDO07     | Error processing for the receipt of IDocs: MDE for IDocs: WMMBXY, WMINVE, WMTORD together |      |
| MWMIDO08     | Message WMMBXY (goods movement) inbound                                                   |      |
| MWMIDO13     | Message WMMBXY: follow-up actions after posting goods                                     |      |



## 6 MYERPRCK - Program Parameters

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### Purpose

Use the upload program myerprck.exe/out to create confirmations/uploads to higher-level systems. In addition to the settings you make directly in the applications, you can also use program parameters to control confirmations/uploads.

### Integration

The confirmation/upload is integrated with numerous components, for example:

- Shop floor data collection
- Tracking and tracing as well as material and production logistics
- Detailed scheduling

Available program parameters:

| Parameters                                | Meaning/use                                                                                                                                                                                                    | Relevant interfaces                                      | Productive release      |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|-------------------------|
| Program parameters to control processing: |                                                                                                                                                                                                                |                                                          |                         |
| /MESTYP=XXXX                              | The parameter MESTYP defines the structure to be generated.                                                                                                                                                    | All                                                      | Yes                     |
| /GRP=XXXX                                 | The grouping type specifies the criterion by which uploads should be grouped.<br>Possible values:<br><br>PLANT --> Groups by plant                                                                             | Requires customizations                                  | Requires customizations |
| /V=sssss                                  | Since SAP R/3 PP does not support correction postings, HYDRA allows to retain confirmations/uploads for correction purposes in HYDRA for a specific period of time.<br><br>Use the parameter /V=sssss (sssss = | EIS-ERP<br><br>EIS-XPPS<br><br>SAP-PPDC<br><br>SAP-PPREM | Yes                     |

| Parameters                                                                                                                                          | Meaning/use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Relevant interfaces                                                                                                                | Productive release |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------|
|                                                                                                                                                     | <p>delay time in seconds) to activate the above described delay when the upload program is called.</p> <p>Examples:</p> <pre>myerprck.exe/out /V=3600</pre> <p>The system only uploads postings that are older than one hour.</p>                                                                                                                                                                                                                                                                                                                                         | <p>SAP-PPPI</p> <p>SAP-PMCC3</p> <p>SAP-PSCC4</p> <p>SAP-COILV</p>                                                                 |                    |
| <p><b>/BIS=DDMMYYHHMM</b></p> <p><b>/BIS=HHMM</b></p> <p><b>/TILLDATE=MM/DD/YYYY</b></p> <p><b>/TILLTIME=sec</b>      <b>after</b><br/>midnight</p> | <p>Use the parameter <b>/BIS=DDMMYYHHMM</b> (date + time) when calling the upload program to enter the delay as a point in time. You can enter this point in time with date and time or you can just enter the time in the format "HHMM". In the latter case, the time refers to the current day.</p> <pre>Myerprck.exe /BIS=2505110600</pre> <p>This parameter uploads postings that were recorded until 06:00 a.m. on 25 May 2011.</p> <pre>Myerprck.exe /BIS=0600</pre> <p>This parameter uploads postings that were recorded until 06:00 a.m. of the current day.</p> | <p>EIS-ERP</p> <p>EIS-XPPS</p> <p>SAP-PPDC</p> <p>SAP-PPREM</p> <p>SAP-PPPI</p> <p>SAP-PMCC3</p> <p>SAP-PSCC4</p> <p>SAP-COILV</p> | Yes                |
| <b>/TZ=+/-sssss</b>                                                                                                                                 | <p>Use the parameter <b>/TZ=+/-sssss</b> to adapt uploads to different time zones. The parameter adjusts the time specifications entered in the fields EXEC__START_TIME, EXEC_FIN_TIME and LOGTIME of the upload structure of the SAP-PPDC interface according to its specifications.</p>                                                                                                                                                                                                                                                                                 | SAP-PPDC                                                                                                                           | Yes                |

| Parameters | Meaning/use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Relevant interfaces                                                                             | Productive release |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------|
| /KST=XXX   | <p>Use this parameter to restrict the data to be uploaded. In this case, the system only uploads data of a specified cost center.</p> <p>Use the parameter /KST=XXX (XXX = cost center, a max. of 8 characters) when calling the upload program myerprck.exe/out to enable the above-described restriction. Then the system only uploads data records that were posted to machines of the specified cost center. The system checks the cost center of the machine/workplace that is entered as the posting workplace/machine in the posting record. The system only checks the cost center of the workplace/machine.</p> <p>You can specify the parameter several times per call.</p> <p><u>Example:</u></p> <pre>Myerprck.exe /KST=BDE100 /KST=BDE200</pre> <p>The system only uploads records that were posted onto machines of the cost center BDE100/BDE200.</p> | EIS-ERP<br>ESI-XPPS<br>SAP-PPDC<br>SAP-PPREM<br>SAP-PPPI<br>SAP-PMCC3<br>SAP-PSCC4<br>SAP-COILV | Yes                |
| /CLEAR_RES | Use the parameter "/CLEAR_RES" to assign an "X" to the field CLEAR_RES of the upload structure when it comes to a final confirmation/upload (record type L40). Consequently, SAP will clear open reservations for the respective order.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | SAP-PPDC                                                                                        | Yes                |
| /NEG_MENGE | By default, quantities (L20/L40) cannot                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | SAP-PPDC                                                                                        | Yes                |

| Parameters      | Meaning/use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Relevant interfaces      | Productive release |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------|
|                 | <p>be uploaded to SAP PP using partial confirmations/uploads via the SAP-PPPDCC interface if data is collected at the same time via the total quantity counter of MDE machines, since SAP is not able to process negative quantities. This type of collection can result in negative quantity postings for yield when OPs are finished.</p> <p>This restriction does no longer apply, if it is possible to process such negative postings (e.g. by using the SAP standard BAPI or customizations). In this case, you can use the program parameter /NEG_MENGE to enable the upload of these quantities.</p> |                          |                    |
| /LA_MNR         | <p>The SAP_PMCC3 interface requires the activity type to be uploaded to SAP PM.</p> <p>The activity type can be identified via the machine/workplace where the posting was performed. Use this program parameter to enable identification of the activity type.</p> <p>Then the system uses the machine to identify the activity type from the <a href="#">activity types</a> kept in HYDRA.</p>                                                                                                                                                                                                            | SAP-PMCC3                | Yes                |
| /IDENT_PRAEFIX= | <p>In the upload structure of the SAP-PPPDCC interface, the field EX_IDENT uniquely identifies uploads from subsystems. HYDRA populates the field.</p> <p>You can add a prefix to the EX_IDENT field to differentiate between uploads</p>                                                                                                                                                                                                                                                                                                                                                                   | SAP-PPPDCC<br>SAP-PPPDCC | Yes                |

| Parameters  | Meaning/use                                                                                                                                                                                                                               | Relevant interfaces                                                                             | Productive release |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------|
|             | <p>from various HYDRA subsystems connected to one SAP instance.</p> <p><u>Example:</u></p> <pre>Myerprck.exe /IDENT_PRAEFIX=ABC</pre> <p>The prefix may only include hexadecimal characters: A –H und 0 – 9.</p>                          |                                                                                                 |                    |
| /ABZEICH=XX | <p>While customizing the order type, you can specify that only signed data records are uploaded.</p> <p>Use the parameter /ABZEICH=XX to specify a period of time in days after that you can upload even unsigned data records.</p>       | EIS-ERP<br>EIS-XPPS<br>SAP-PPDC<br>SAP-PPREM<br>SAP-PPPI<br>SAP-PMCC3<br>SAP-PSCC4<br>SAP-COILV | Yes                |
| /TRANSFER=  | <p>Use the parameter "/TRANSFER=" to only upload records whose specifications were transferred from a specific system.</p> <p>The transfer indicator is set during HYDRA inbound processing and may vary from interface to interface.</p> | EIS-ERP<br>ESI-XPPS<br>SAP-PPDC<br>SAP-PPREM<br>SAP-PPPI<br>SAP-PMCC3<br>SAP-PSCC4<br>SAP-COILV | Yes                |

| Parameters       | Meaning/use                                                                                                                                                                                                                                                                      | Relevant interfaces                                                                             | Productive release |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------|
| /NOTTRANSFER=XXX | <p>Use the parameter "/NOTTRANSFER=" to only upload records whose specifications were NOT transferred from a specific system.</p> <p>The transfer indicator is set during HYDRA inbound processing and may vary from interface to interface.</p>                                 | EIS-ERP<br>ESI-XPPS<br>SAP-PPDC<br>SAP-PPREM<br>SAP-PPPI<br>SAP-PMCC3<br>SAP-PSCC4<br>SAP-COILV | Yes                |
| /SEK             | <p>The EIS-ERP interface uploads the times of resource performance accounts in hours.</p> <p>In particular with very short lead times this may effect that logon times are cut off by a conversion into hours.</p> <p>Use this program parameter to upload times in seconds.</p> | EIS-ERP<br>ESI-XPPS                                                                             | Yes                |
| /RMTYP=          | <p>When customizing the order type, you can assign an upload type to the order type.</p> <p>Use this program parameter to only upload data records of this upload type.</p> <p>You can specify the parameter several times per call.</p>                                         | EIS-ERP<br>ESI-XPPS<br>SAP-PPDC<br>SAP-PPREM<br>SAP-PPPI<br>SAP-PMCC3<br>SAP-PSCC4<br>SAP-COILV | Yes                |

| Parameters | Meaning/use                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Relevant interfaces                                                                             | Productive release      |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------|
| /KAT=      | <p>When customizing the order type, you can connect the order type with a category.</p> <p>Use the program parameter /KAT= to only upload data records of this category.</p> <p>You can specify the parameter several times per call.</p>                                                                                                                                                                                                                          | EIS-ERP<br>ESI-XPPS<br>SAP-PPDC<br>SAP-PPREM<br>SAP-PPPI<br>SAP-PMCC3<br>SAP-PSCC4<br>SAP-COILV | Yes                     |
| /SART=     | <p>The system only uploads ADE log postings of the specified record type.</p> <p>Therefore, you can use different program parameters per call and record type for uploading.</p> <p>Requirement: You have to activate the corresponding uploads when customizing the order type.</p> <p>You can specify the parameter several times per call.</p> <p><u>Example:</u></p> <pre>Myerprck.exe /SART=A /SART=E</pre> <p>⇒ The system only uploads A and E records.</p> | EIS-ERP<br>ESI-XPPS<br>SAP-PPDC<br>SAP-PPREM<br>SAP-PPPI<br>SAP-PMCC3<br>SAP-PSCC4<br>SAP-COILV | Yes                     |
| /NOLOCK    | <p>When starting the upload program, the system checks if there are any lock entries for the database table ADE_PROTOKOLL. If this is the case,</p>                                                                                                                                                                                                                                                                                                                | All                                                                                             | Requires customizations |

| Parameters    | Meaning/use                                                                                                                                                                                                                                                                                                                                                                                                                        | Relevant interfaces               | Productive release |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|--------------------|
|               | <p>the upload is not carried out.</p> <p>You can use this program parameter to prevent this check.</p> <p>Set this parameter, in particular, if the upload is not based on the table ade_protokoll.</p>                                                                                                                                                                                                                            |                                   |                    |
| /EINH_CC34    | <p>The interfaces SAP-PMCC3 and SAP-PSCC4 transfer the uploaded activity quantity in seconds (SEC) to SAP. Use the parameter <b>"/EINH_CC34"</b> to upload the data in other units. The following units are supported:</p> <p>Hours: H, HUR, STD</p> <p>Minutes: MIN</p> <p>Seconds: SEC</p> <p><u>Example:</u></p> <pre>Myerprck.exe /EINH_CC34=HUR</pre> <p>The system uploads the recorded times in the unit "HUR" (hours).</p> | <p>SAP-PMCC3</p> <p>SAP-PSCC4</p> | Yes                |
| /SDAT_STORNO  | <p>The SAP-PPPDCC interface transfers the change date along with the correction records.</p> <p>Use this program parameter to upload the initially collected shift date instead.</p>                                                                                                                                                                                                                                               | SAP-PPPDCC                        | Yes                |
| /NORFC_STORNO | <p>The SAP-PPPDCC interface transfers the cancellation records via <a href="#">sRFC</a>.</p> <p>Use the program parameter to transfer the data in the IDoc format to SAP. To do</p>                                                                                                                                                                                                                                                | SAP-PPPDCC                        | Yes                |



| Parameters       | Meaning/use                                                                                                                                                                                                                                                                                                     | Relevant interfaces                               | Productive release      |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|-------------------------|
|                  | <p>so, inbound processing must be implemented in SAP.</p> <p>The system uploads the cancellation records via the standard PP-PDC segment (with record type K20/K40) as if the PP-PDCC license was not available.</p>                                                                                            |                                                   |                         |
| /PI              | <p>If you use the SAP Process Integration (previously: Exchange Infrastructure) to communicate with SAP, the version of the transferred segment is checked more strictly.</p> <p>Use the program parameter to transfer segment names with the version number (i.e. the trailing zeros of the segment name).</p> | <p>SAP-PPDC</p> <p>SAP-PMCC3</p> <p>SAP-PSCC4</p> | Yes                     |
| /INDEX_TMP_TABLE | <p>Use this parameter to accelerate uploads if ORACLE is used as database system and large amounts of data are affected.</p> <p>To do so, use an index for a temporary table where all data to be uploaded is transferred in a first step.</p>                                                                  | All                                               | Requires customizations |
| /UE_PARAMS=      | Program parameter for the stand-alone user exit processing (DD format).                                                                                                                                                                                                                                         | Various                                           | Yes                     |
| /NOSTORNO        | <p>Use this program parameter to prevent cancellation records from being uploaded.</p> <p>Therefore, you can use different program parameters per call and record type for uploading.</p> <p>Requirement: You have to activate the</p>                                                                          | All                                               | Yes                     |

| Parameters                                          | Meaning/use                                                                                                                                                                                                                                                                                                                                                                                                                                | Relevant interfaces | Productive release      |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------|
|                                                     | corresponding uploads when customizing the order type.                                                                                                                                                                                                                                                                                                                                                                                     |                     |                         |
| /RECALC_NEG_YIELD                                   | Use this parameter to offset negative yield with already posted positive uploads.                                                                                                                                                                                                                                                                                                                                                          | SAP-PPPDCC          | Requires customizations |
| Program parameters to use the SIGUSR communication: |                                                                                                                                                                                                                                                                                                                                                                                                                                            |                     |                         |
| /LOGGING                                            | Use this program parameter to activate communication from the database table HYD_LOGGING.<br><br>To do so, a customization might be required.                                                                                                                                                                                                                                                                                              | INDIVIDUAL CASE     | Yes                     |
| /WAIT_SIGUSR1=XX                                    | The program parameter specifies the time in seconds that has to pass before the upload is performed via the SIGUSR communication even without trigger.                                                                                                                                                                                                                                                                                     | INDIVIDUAL CASE     | Yes                     |
| /PEEK_SIGUSR1=XX                                    | Use this parameter to delay execution of an action triggered by the SIGUSR communication.<br><br>The delay time is entered in seconds for this parameter.<br><br>The program interprets this time as follows:<br><br>If within the next second after the initial trigger there is another trigger, then wait for not more than <specified value> seconds.<br><br>If in a specific case, triggers would indeed arrive every second then the | INDIVIDUAL CASE     | Yes                     |

| Parameters        | Meaning/use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Relevant interfaces | Productive release |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--------------------|
|                   | WAIT_SIGUSR time (e.g. 120 seconds) would apply; i.e. the system would in fact perform the upload after 2 minutes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                     |                    |
| /SEND_SIGUSR1=    | <p>This program parameter defines which other process/ program must be triggered after processing by the SIGUSR communication.</p> <p>Specify the process/program WITHOUT file extension.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | INDIVIDUAL CASE     | Yes                |
| /COUNT_SIGUSR1=XX | <p>Uploading in signal mode can hardly be subjected to tracing. This is due to the fact that the program in those cases is started once via the scheduler but won't shut off. Any redirection of the program call with -d to a log file will then necessarily lead to very large log files, which will negatively affect the performance.</p> <p>Use the new program parameter /COUNT_SIGUSR1=XX to specify after how many calls the program will automatically shut down. A call in these instances is both, a call via SIGUSR communication and the cyclical program execution which is controlled via the parameter /WAIT_SIGUSR1.</p> <p>Then the scheduler restarts the program. But this will lead to a time period "t" during which SIGUSR calls will not be processed. It is, however, assumed that this will not lead to data losses since the data to be uploaded are already saved to</p> | INDIVIDUAL CASE     | Yes                |

| Parameters                                                            | Meaning/use                                                                                                                                                                                                                                                   | Relevant interfaces | Productive release |
|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--------------------|
|                                                                       | <p>the DB.</p> <p>Benefits:</p> <p>If the program is started via a script (*.scr) from the scheduler, you can store there the routine to generate a date/ time stamp file name for the log file to be created. This allows to restrict the log file size.</p> |                     |                    |
| Program parameters for debugging/ tracing/ testing/ logging purposes: |                                                                                                                                                                                                                                                               |                     |                    |
| /ONLYERR                                                              | <p>This program parameter specifies that system log entries are only created if an error occurred during uploading.</p> <p>This reduces the entries in the system log.</p>                                                                                    | All                 | Yes                |
| /SIM                                                                  | The system does not upload/confirm data during simulations (the uploaded/confirmed indicator is set to "True").                                                                                                                                               | All                 | No                 |
| /SIMULATION                                                           | The system does not upload/confirm data to SAP during simulation (confirmed/uploaded indicator will not be changed).                                                                                                                                          | All                 | No                 |